

A multiscale physiologically based pharmacokinetic model to support mRNA-encoded BiTE therapy in cancer treatment

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In the field of cancer therapy, bispecific T cell engagers (BiTEs) have demonstrated significant potential. However, their clinical application is constrained by challenges in production and limited plasma half-life. *In vitro*-transcribed (IVT) mRNA formulations emerge as a promising alternative, offering adaptability and cost-efficiency. Yet, the intricate relationship between mRNA dosage, antibody production, and the distribution of mRNA and proteins requires a deeper understanding. To address these issues, we present a novel physiologically based pharmacokinetic (PBPK) model to characterize the pharmacokinetics of BiTEs. This model predicts the distribution patterns of both recombinant and mRNA-encoded BiTEs by extending an established PBPK model with a hierarchical multiscale framework calibrated and validated using preclinical data from existing literature. The extended PBPK model can be adapted to various mRNA-based therapeutic formulations, facilitating *in-silico* exploration of different drug administration scenarios. It can provide valuable support for optimizing dose and schedule and allows the efficient investigation of drug distribution at a whole-body scale. This approach promises to enhance the personalization and effectiveness of cancer therapies, reduce research time and costs, and significantly advance the development of mRNA-based BiTEs for cancer treatment.

INTRODUCTION

As of 2024, cancer continues to pose a significant global health challenge. In the United States alone, the year 2023 has seen nearly 2 million new cancer cases and over 600 thousand cancer-related deaths.¹ These statistics underscore the pressing need to advance therapeutic strategies that can effectively reduce the impact of this widespread disease, such as immunotherapies that harness the body's innate defense mechanisms to combat cancer.

Bispecific T cell engagers (BiTEs) represent a highly promising immunotherapeutic strategy, functioning by concurrently binding to a tumor-associated antigen on cancer cells and a surface molecule (typi-

cally CD3) on T cells, thereby inducing tumor lysis.² BiTEs act as molecular bridges between T cells and cancer cells, facilitating the close proximity necessary for enhanced T cell activation, which in turn triggers the release of cytokines and cytotoxic molecules responsible for tumor cell destruction.³ BiTE therapies have demonstrated significant efficacy in treating several B cell-associated hematologic malignancies.^{4–6}

However, the therapeutic potential of BiTEs is constrained by challenges such as on-target off-tumor toxicity, short half-life, cytokine release syndrome, and immunogenicity, which collectively narrow their therapeutic window.^{7–10} These limitations could be partially addressed by employing mRNA-encoded BiTEs, wherein mRNA is transcribed by the host cells into bispecific antibodies.¹¹ mRNA-based BiTEs offer extended half-life and the capacity for multiplexing—i.e., co-delivery of two different mRNAs encoding complementary BiTEs. This innovative approach provides improved precision, prolonged pharmacokinetics, and greater adaptability in cancer immunotherapy compared to conventional BiTEs.^{12,13}

mRNA-encoded BiTEs are a rather recent technology, warranting further investigation in determining optimal doses and dosing schedules, as well as elucidating the intricate processes governing mRNA and antibody trafficking. Precise prediction based on rodent preclinical findings paves the way to study *in silico* larger animal models and even humans, greatly speeding up the advancement of BiTE therapies to eventually untangle complex clinical scenarios.

In response to these challenges, our work aims to develop a robust computational tool founded upon the principles of the

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physiologically based pharmacokinetic (PBPK) modeling. PBPK models have long been recognized for their efficacy in simulating the distribution and behavior of pharmaceutical agents within the human body,^{14,15} and they are well established and widely recognized as valuable tools in drug development by regulatory agencies such as the FDA and EMA.¹⁶ Within the context of mRNA-encoded BiTE therapies, we extend the seminal work of Li and Shah (2019),¹⁷ adapting their model of monoclonal antibodies to simulate BiTE distribution within tissues by enhancing its suitability for the unique characteristics of mRNA-encoded bispecific antibodies. Our model further incorporates a phenomenological description of the intricate processes of mRNA trafficking, translation, and the subsequent production of functional bispecific antibodies. This phenomenological layer is concatenated with the PBPK model for the description of the trafficking of the antibodies. Altogether, this model empowers us to make precise predictions regarding the distribution and behavior of mRNA-encoded BiTEs at a whole-body scale.

Notably, the developed model has the ability to predict scenarios encompassing both the administration of recombinant BiTEs and those synthesized through mRNA injection. The model is also designed to accommodate new equations to account for alternative administration sites, such as oral, intravenous (i.v.), transdermal, and others, to aid *in-silico* exploration of the most efficient and effective delivery methods. Moreover, it can contribute to assessing the impact of dosing frequency, timing, and duration on drug exposure, allowing for the design of dosing schedules that maximize the therapy efficacy.

RESULTS

Model definition

Our study endeavors to extend previous investigations of protein transport in preclinical species by incorporating exogenous mRNA-encoded antibodies within the framework of PBPK modeling. PBPK models offer a comprehensive quantitative analysis of drug distribution in both plasma and tissue compartments, owing to their intricate structure. Additionally, their applicability across species, facilitated by a shared anatomical framework among mammalian organisms, confers significant advantages in translational research across various animal models and aids in the design of clinical trials.

In our investigation, we employed a modeling framework based on the mouse preclinical model by leveraging the one proposed by Li and Shah (2019)¹⁷ to simulate protein distribution, designated in this work as the protein transport module (PTM). Furthermore, we expanded upon this framework by integrating a simplified PBPK model describing mRNA transport and subsequent protein production, termed the mRNA-transport module (MTM). Specifically, with respect to the work presented in Li and Shah (2019),¹⁷ we added a representation of the dynamics of LNP-mRNA uptake in the liver and the subsequent antibody production; a detailed description of the equations is provided in the following section.

This modular approach facilitates the comparison of mRNA-encoded therapeutics to recombinant ones, enabling the utilization of data pertaining to recombinant therapies to evaluate their mRNA-encoded counterparts.

Briefly, the PTM aims to model the physiological dynamics of the entire body, encompassing 16 specific tissues interconnected through blood and lymphatic circulation. Each tissue compartment is further subdivided into plasma, endosomal, and interstitial sub-compartments, with protein molecules presumed to enter via arterial plasma flow and exit via venous plasma flow. Lymph flow facilitates fluid-phase convection from vascular to interstitial spaces, cycling through the lymphatic system. Protein disposition at the tissue level in our proposed PBPK model is characterized using the two-pore hypothesis, with additional parameter details available in Li and Shah (2019).¹⁷ A schematic representation of the PTM is represented in Figure 1.

Since the presented model serves as a proof of concept of the effectiveness of PBPK modeling in improving our knowledge around mRNA-based therapeutics in general, we focused on the protein trafficking occurring during mRNA-encoded bispecific antibodies, which did not include the neonatal Fc receptor (FcRn) sequence. FcRn is often employed for extending the half-life and enhancing the efficacy of IgG-like antibodies by protecting them from degradation.^{18,19} This deliberate omission ensures a more parsimonious and accurate representation of the pharmacokinetics for the therapy in question, reflecting its unique characteristics and optimizing model performance.

Lipid nanoparticles (LNPs) currently serve as the most popular delivery system for mRNA-based products, including mRNA-encoded antibodies.²⁰ Unmodified LNPs are naturally taken up by the liver, even though new formulations are under development to cf. them different tropism and organ-selectiveness.²¹ In our model, the MTM has been constructed to comply with the product-specific characteristics elucidated in Huang et al. (2023),²² which stem from the encapsulation of mRNA encoding B7H3×CD3 BiTE within a novel liver-targeted ionizable LNP delivery system, intravenously administered.

According to the referenced data, the transfection efficacy was the highest in liver tissue (about 99%, as reported in Huang et al., 2023²²), followed by spleen tissue, while dropping significantly in all the other studied organs. Thus, by qualitatively assessing the contribution of each organ during the transfection process, our model structure considers exclusively the injection site (blood) and the target site for mRNA (liver), where protein production takes place. Specifically, our model assumes that the LNP-mRNA therapeutic is administered intravenously, allowing it to reach the liver, where it is taken up by hepatocytes, ultimately leading to antibody production. A schematic representation of this simplified biological framework, which forms the basis of our model, is shown in Figure 2.

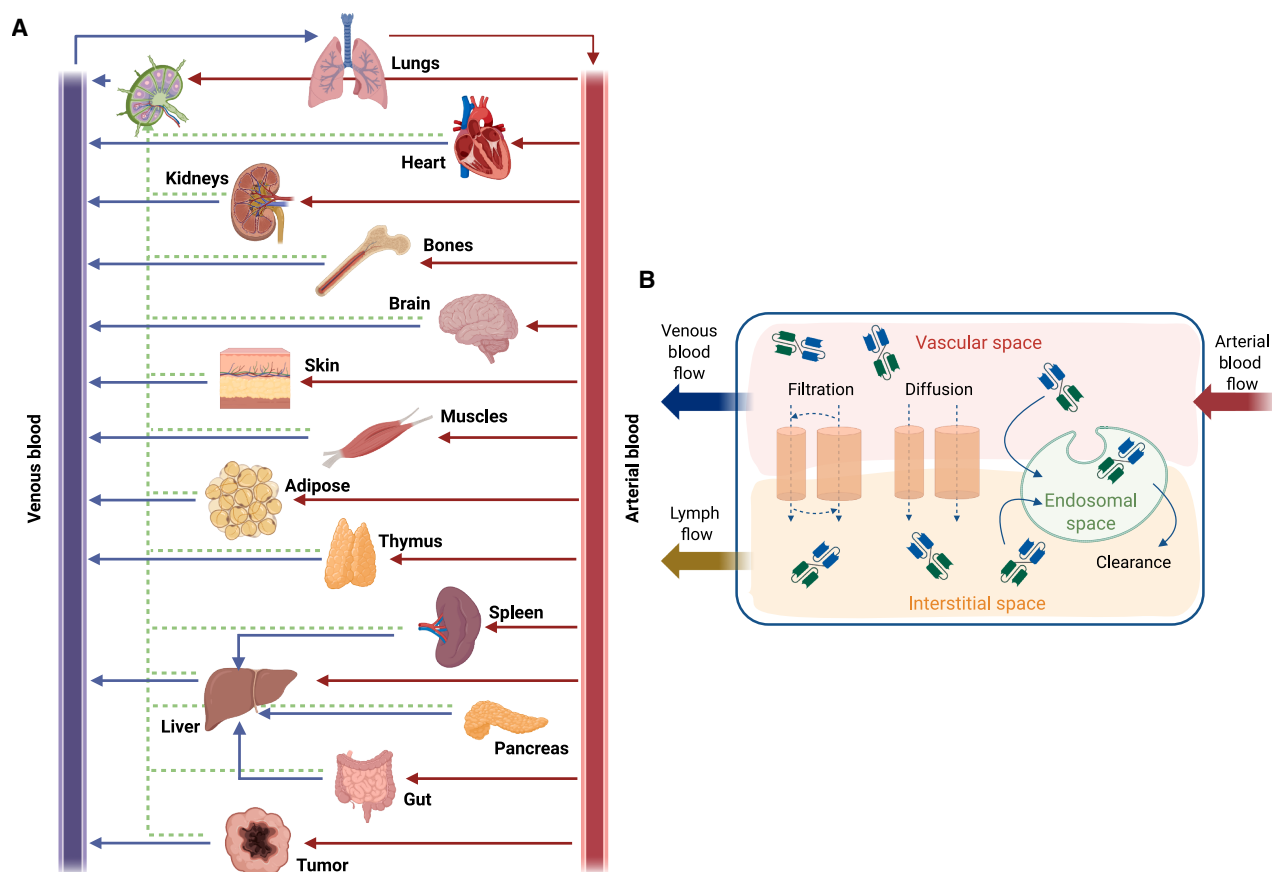


Figure 1. Schematic representation of the protein transport module

Figure adapted from Li and Shah (2019)¹⁷ to the specific case of BiTE's trafficking. (A) represents the whole-body structure of the PBPK model. All organs are connected in an anatomical manner with blood flow (solid arrows) and lymphatic flow (dashed arrows). Arrows represent the direction of the flow. While venous blood is a compartment in our model, arterial blood has been represented via fluxes only. Each tissue within this model, except blood and lymph node, is divided into three sub-compartments: vascular, endosomal, and interstitial space, as shown in (B). (B) schematically represents the structure of the tissue level PBPK model. Protein disposition at the tissue level in our PBPK model is described using the two-pore hypothesis, with additional parameter details available in Li and Shah (2019).¹⁷ Created in BioRender. Marchetti, L. (2025) <https://BioRender.com/n17j928>.

mRNA metabolism is an extremely regulated and complex process, therefore, relying solely on the mRNA transferred to the liver as the primary variable would not accurately fit with the experimental data. The rise in plasma antibody concentration, triggered by mRNA translation in hepatocytes, occurs with a certain delay that needs to be reflected in the model by either describing in great detail the mechanistic pathways involved in the metabolism of exogenous mRNA *in vivo* or by providing a phenomenological description of the events. We selected the latter approach, defining a parsimonious model description that, importantly, ensures parameter identifiability.

Adhering to the classical approach employed in mathematical modeling when a delay is expected, we introduced the minimum number of variables required to maintain the model parsimonious, by assuming two sequential stages of mRNA processing within the liver. Termed the short-chain model (SCM), this conceptual frame-

work (Figure 3) considers the liver as harboring mRNA in two distinct states, here indicated as $mRNA_{LV1}$ and $mRNA_{LV2}$, both contributing to the generation of BiTEs at different translation rates—one for each state—alongside a transition rate between states, encapsulating the liver's temporal responsiveness. The chain of events described can be summarized as follows: (1) the mRNA is injected intravenously; (2) it is then transferred through the circulatory system to the liver in quantities proportional to those reported in the experimental data; and (3) upon arrival at the liver, part of the mRNA is transcribed (and degraded) while the rest is transferred to its second state, where it undergoes the same process. Specifically, the degradation term introduced in the model represents the transient nature of mRNA, which is rapidly degraded by RNases—enzymes responsible for breaking down RNA molecules. These RNases catalyze the hydrolysis of the mRNA, thus ensuring the mRNA's transient presence in the system and preventing its accumulation over time. This can be described

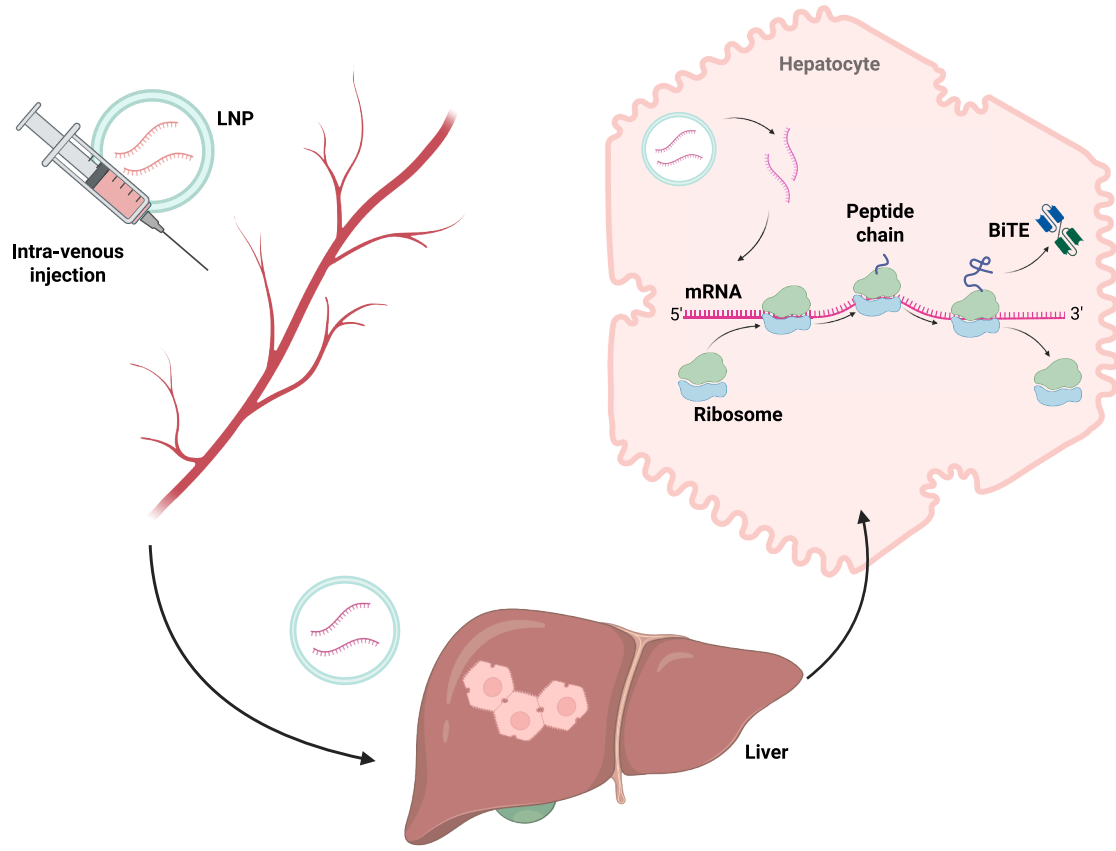


Figure 2. Schematic representation of the mRNA-transport Module

Following intravenous administration of LNP-mRNA, the majority of LNPs migrate to the liver, get up-taken by hepatocytes and release mRNA into their cytosol, where translation of BiTEs occurs. To play their role, BiTEs need to be trafficked through blood at a whole-body scale, which is modeled by the PTM. Created in BioRender. Marchetti, L. (2025) <https://BioRender.com/h11n272>.

mathematically by the following ordinary differential equation (ODE) system:

$$\frac{d \text{mRNA}_{BL}}{dt} = -k_{BL2LV} \text{mRNA}_{BL} - k_d^{\text{mRNA}} \text{mRNA}_{BL} \quad (\text{Equation 1})$$

$$\begin{aligned} \frac{d \text{mRNA}_{LV1}}{dt} = & +k_{BL2LV} (1 - r_{\text{others}}^{\text{mRNA}}) \text{mRNA}_{BL} \left(\frac{V_{BL}^{\text{TOT}}}{V_{LV}^{\text{IS}}} \right) \\ & - k_{s12} \text{mRNA}_{LV1} - k_d^{\text{mRNA}} \text{mRNA}_{LV1} \end{aligned} \quad (\text{Equation 2})$$

$$\frac{d \text{mRNA}_{LV2}}{dt} = +k_{s12} \text{mRNA}_{LV1} - k_d^{\text{mRNA}} \text{mRNA}_{LV2} \quad (\text{Equation 3})$$

where mRNA_{BL} , mRNA_{LV1} , and mRNA_{LV2} represent, respectively, the mRNA concentration in the blood and in the liver (at the two different states); k_{BL2LV} is the transfer rate from blood to liver; $r_{\text{others}}^{\text{mRNA}}$ is the ratio of the mRNA injected that has been transferred to all the other organs beside the liver; k_{s12} is the phenomenological

transition rate from the first to the second state of the mRNA in the liver; and, finally, k_d^{mRNA} is the degradation rate of the mRNA.

In our model, we hypothesize that the translation process takes place in the interstitial sub-compartment of the liver, where we assume hepatocytes, the main actors of LNP-mRNA uptake and translation,²³ are found.²³ Therefore, we revise the dynamics of liver proteins outlined in Li and Shah (2019),¹⁷ in light of the proteins that are now synthesized during exogenous mRNA translation. Following the same notation of the fifth equation in Li and Shah (2019),¹⁷ the equation representing the evolution of liver antibody concentration in the interstitial space (C_{LV}^{IS}) over time becomes as follows:

$$\begin{aligned} V_{LV}^{\text{IS}} \times \frac{d C_{LV}^{\text{IS}}}{dt} = & CL_{tp,LV} \times C_{LV}^V - (1 - \sigma_{LV}^{\text{IS}}) \times J_{LV} \times C_{LV}^{\text{IS}} \\ & - CL_{up} \times C_{LV}^{\text{IS}} + k_{tr}^{\text{mRNA}_{LV1}} \times \text{mRNA}_{LV1} \times V_{LV}^{\text{IS}} \\ & + k_{tr}^{\text{mRNA}_{LV2}} \times \text{mRNA}_{LV2} \times V_{LV}^{\text{IS}} \end{aligned} \quad (\text{Equation 4})$$

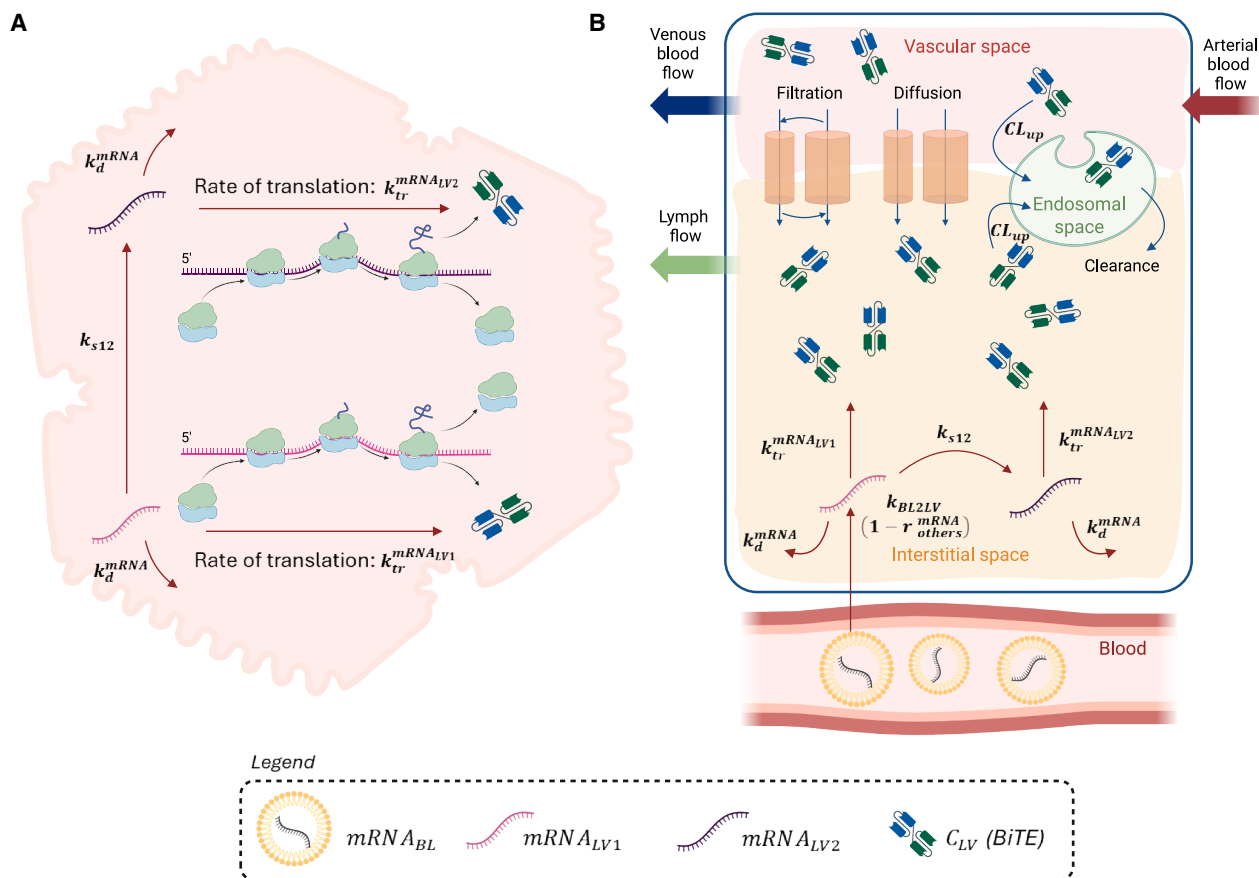


Figure 3. Schematic representation of mRNA translation in the short-chain model

(A) represents the set of reactions inside hepatocytes. Two mRNA stages are present to consider the delay expected in the protein synthesis process. Additionally, two different velocities of translation are considered to encapsulate the liver's temporal responsiveness. (B) represents the tissue level dynamics modeled in the liver. The mRNA injected into blood is transferred to the liver and its translation takes place in the interstitial space, following the SCM. When BiTEs are produced, they are then transported at the whole-body scale via the PTM. Created in BioRender. Marchetti, L. (2025) <https://BioRender.com/x87j373>.

where $CL_{tp,LV}$ represents the uptake clearance of protein molecules mediated by the *trans*-capillary exchange processes (defined as the “two-pore clearance” rate in Sepp et al. [2015]²⁴), σ_{LV}^{IS} is the interstitial reflection coefficient, as a measure of the level of resistance at which proteins are subjected along the lymphatic system transport, J_{LV} is the lymph flow (set at 0.2% of the plasma flow to the liver), CL_{up} is the rate at which proteins in the interstitial space are taken up by the vascular endothelial cells via fluid-phase pinocytosis, and $k_{tr}^{mRNA_{LV1}}$ for $i = 1, 2$ are the rates at which BiTEs are produced via translation in each mRNA phase.

The trafficking of the translated therapeutics is then computed by relying on the PTM according to the two-pore PBPK model from Li and Shah (2019)¹⁷ to which we refer for any further detail on parameter estimates and equation descriptions of the original model.

Model training and validation

As detailed in the [materials and methods](#) section, the model training utilized pharmacokinetic (PK) data from the recombinant protein administration to parameterize the PTM,¹⁷ accounting for the physiochemical properties of the molecule while fitting the experimental data. Compared to the original parameterization reported in Li and Shah (2019),¹⁷ we re-estimated the parameter for the rate of pinocytosis, the process where the cell membrane engulfs external fluids and small molecules to release them into the cytosol.²⁵ Once the pinocytosis-related parameter was estimated, it was integrated into the default parameterization for the PTM. This allowed us to estimate the parameters for the MTM. Model training used time series data of serum antibody concentrations in mice following a single i.v. dose of the mRNA-based therapeutic, while model validation involved evaluating three alternative mRNA-based doses, as further described in the [materials and methods](#) section.

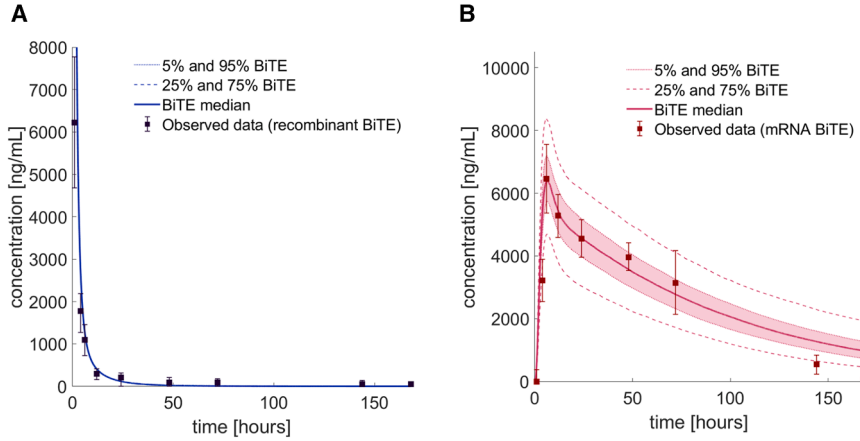


Figure 4. Training results for the short-chain model

Model training results on plasma PK time series of the recombinant BiTEs concentration (A) and the mRNA-encoded one (B), the experimental data are the ones reported in Huang et al. (2023). Uncertainty of the mRNA-based output is estimated by performing a virtual population. Solid lines represent the median concentration of BiTEs in plasma, dot lines represent the first and the third quartile and dashed lines represent the 5% and 95% of the estimated values in the two experimental settings, respectively.

As anticipated in the [model definition](#) section, our extended model introduces six new parameters into the MTM to simulate mRNA trafficking and translation: the rate of mRNA degradation, the rate of mRNA transfer from plasma to the liver, the ratio of mRNA transferred to organs other than the liver, the mRNA transfer rate within the liver, and the two mRNA efficacy rates in protein translation. The parameter for the proportion of mRNA released in organs other than the liver was set to 0.011 based on Huang et al. (2023),²² indicating that approximately 99% of administered mRNA accumulates in the liver after 6 h.

The model was trained using published concentration time-profile data on both recombinant BiTEs and mRNA-encoded BiTEs from Huang et al. (2023)²² (Figure 4). The initial BiTE concentration in all modeled sub-compartments was set to zero except for plasma, which reflected the injected amounts reported by Huang et al. (2023)²²: for recombinant BiTE administration, C_{PLASMA} was initialized to account for the 6 mg/kg injected dose in the model compartment volume, while for the mRNA-encoded protein C_{PLASMA} was set to reflect a 1.5 mg/kg administration. The model fit and parameter estimates from the training phase are shown in Figure 4 and Table 1, respectively. In Figure 4, the uncertainty in the output (indicated by the light-red shaded region) was estimated by conducting a virtual population study on the newly introduced mRNA parameters and the endothelial pinocytosis rate, as specified in the [materials and methods](#) section.

The model training results (Figure 4) show that our platform can effectively reproduce the experimental data for both the recombinant protein and the mRNA-based formulation. Furthermore, the estimate obtained for the mRNA's degradation rate by the fitting is consistent with experimentally estimated ranges since the corresponding average half-life for the whole-body mRNA concentration resulted in around 45 h, in agreement with the order of magnitude reported in the literature.²⁶ However, it's worth noting that our model slightly overestimates antibody concentrations in the latter part of the observed time frame.

To validate our approach, we tested the ability of our model to predict the PK in plasma of the mRNA-BiTE at increasing doses, by reproducing *in silico* the dose escalation experiment reported in Huang

et al. (2023),²² where the authors reported the serum concentration of BiTE in mouse 24 h after administering increasing amounts (0, 0.5, 1, 1.5, and 2 mg/kg). Notably, our platform predicted the plasma antibody concentrations 24 h after administration within the variances of the experimental data in all cases (Figure 5).

Model refinement toward a product-specific platform

The overestimation of therapeutic availability in plasma computed by our model toward the end of the simulation, at about the 150th hour mark (Figure 4), juxtaposed with the distinctive PK profile inherent to this specific mRNA-based product, has motivated us to find an alternative and less parsimonious modeling approach for the mRNA translation phase. This endeavor aims to tailor our computational platform to accommodate the specific attributes characterizing the type of BiTE herein described. Notably, an additional phase in the pharmacokinetics of BiTE in blood has been discerned from the data,²² diverging from the classical behavior typically expected in dynamics of this nature. In this regard, in line with the foundational principles of PBPK modeling, we propose an extended modeling framework informed by the number of sign changes observed in the derivative of the experimental data. Specifically, the serum antibody concentration curve exhibits two main peaks, therefore, three distinct changes in slope, suggesting the incorporation of a third mRNA stage within the chain of states taken by mRNA in the liver. We designate this novel formulation of the MTM as the long-chain model (LCM), schematically depicted in Figure 6A.

Consequently, a modification in the mathematical representation of the sequential events is warranted, leading to the following ODE system for the MTM module:

$$\frac{d \text{mRNA}_{BL}}{dt} = -k_{BL2LV} \text{mRNA}_{BL} - k_d^{\text{mRNA}} \text{mRNA}_{BL} \quad (\text{Equation 5})$$

$$\begin{aligned} \frac{d \text{mRNA}_{LV1}}{dt} = & +k_{BL2LV} (1 - r_{\text{others}}^{\text{mRNA}}) \text{mRNA}_{BL} \left(\frac{V_{BL}^{\text{TOT}}}{V_{LV}^{\text{IS}}} \right) \\ & - k_{s12} \text{mRNA}_{LV1} - k_d^{\text{mRNA}} \text{mRNA}_{LV1} \end{aligned} \quad (\text{Equation 6})$$

Table 1. Table of parameter estimates computed for SCM and LCM

Parameter	Description	Estimate	Units
SCM parameters			
k_{BL2LV}	migration rate of mRNA from the vascular system to the liver	6.804×10^{-1}	[1/h]
k_{s12}	transition rate from the first to the second mRNA state in the liver	4.807×10^{-1}	[1/h]
k_d^{mRNA}	mRNA clearance rate	1.08×10^{-2}	[1/h]
$k_{tr}^{mRNA_{LV1}}$	rate of protein production in the liver from mRNA in the first state	9.682×10^{-1}	[1/h]
$k_{tr}^{mRNA_{LV2}}$	rate of protein production in the liver from mRNA in the second state	3.182×10^{-1}	[1/h]
CL_{up}	fluid phase pinocytosis rate	1×10^{-1}	[1/h]
LCM parameters			
k_{BL2LV}	migration rate of mRNA from the vascular system to the liver	1	[1/h]
k_{s12}	transition rate from the first to the second mRNA state in the liver	1.03×10^{-1}	[1/h]
k_{s23}	transition rate from the second to the third mRNA state in the liver	2.44×10^{-2}	[1/h]
k_d^{mRNA}	mRNA clearance rate	2.94×10^{-2}	[1/h]
$k_{tr}^{mRNA_{LV1}}$	rate of protein production in the liver from mRNA in the first state	6.816×10^{-1}	[1/h]
$k_{tr}^{mRNA_{LV2}}$	rate of protein production in the liver from mRNA in the second state	1.01×10^{-2}	[1/h]
$k_{tr}^{mRNA_{LV3}}$	rate of protein production in the liver from mRNA in the third state	1.574	[1/h]
CL_{up}	fluid phase pinocytosis rate	1×10^{-3}	[1/h]

The parameter CL_{up} has been estimated using the PK data of the recombinant version of the product. The remaining parameters have been estimated using the PK data of mRNA-encoded BiTEs. Identifiability analysis confirmed the local identifiability of all model parameters.

$$\frac{d \text{mRNA}_{LV2}}{dt} = +k_{s12} \text{mRNA}_{LV1} - k_{s23} \text{mRNA}_{LV1} - k_d^{mRNA} \text{mRNA}_{LV2} \quad (\text{Equation 7})$$

$$\frac{d \text{mRNA}_{LV3}}{dt} = +k_{s23} \text{mRNA}_{LV2} - k_d^{mRNA} \text{mRNA}_{LV2} \quad (\text{Equation 8})$$

where, in addition to the equations previously introduced for the SCM, mRNA_{LV3} and k_{s23} represent the mRNA concentration in the third state and the phenomenological transition rate from the second to the third state, respectively. Consequently, the dynamics of hepatic BiTE concentration has been modified as follows:

$$\begin{aligned} V_{LV}^{IS} \times \frac{d C_{LV}^{IS}}{dt} = & CL_{tp,LV} \times C_{LV}^V - (1 - \sigma_{LV}^{IS}) \times J_{LV} \times C_{LV}^{IS} \\ & - CL_{up,LV} \times C_{LV}^{IS} + k_{tr}^{mRNA_{LV1}} \times \text{mRNA}_{LV1} \times V_{LV}^{IS} \\ & + k_{tr}^{mRNA_{LV2}} \times \text{mRNA}_{LV2} \times V_{LV}^{IS} \\ & + k_{tr}^{mRNA_{LV3}} \times \text{mRNA}_{LV3} \times V_{LV}^{IS}. \end{aligned} \quad (\text{Equation 9})$$

As previously done for the SCM, the performance of the model during the training phase and the validation results are shown in Figures 6B and 6C, respectively, while the computed parameter estimates are reported in Table 1. This type of modeling performs very well within a 24-h time frame, with validation performance in close agreement with the previous SCM model. Moreover, it exhibits

improved performance in blood clearance, still preserving the local identifiability of model parameters.

DISCUSSION

Both the SCM and the LCM implementations demonstrated their value in capturing the PK dynamics of mRNA-BiTE in blood (Figures 4, 5, and 6). This lays the foundation for leveraging the model as an *in-silico* prediction platform. For example, our model adeptly captures the dynamic behavior of BiTEs across several dosages, highlighting its potential to support the development of mRNA-encoded therapies. In this context, given that both SCM and LCM approaches yield antibody concentrations at 24 h post-administration within the variance of experimental data, a time series for validation would have offered a more comprehensive means to differentiate between the two modeling approaches and enhanced the understanding of their performance.

Furthermore, this multiscale and multi-layer PBPK platform could play a role in predicting the dose-response relationship of mRNA and protein expression, depending on the specific design of the therapeutics (e.g., LNP formulation and tissue targeting). Indeed, despite the multitude of involved processes at each step, such as mRNA uptake, endosomal release, mRNA decomplexation and translation, and mRNA degradation, protein expression follows a linear dose dependence, demonstrating the excellent quantitative control mRNA yields over protein expression.²⁷ However, this linear correlation is sensible to variations in the number of mRNA molecules per

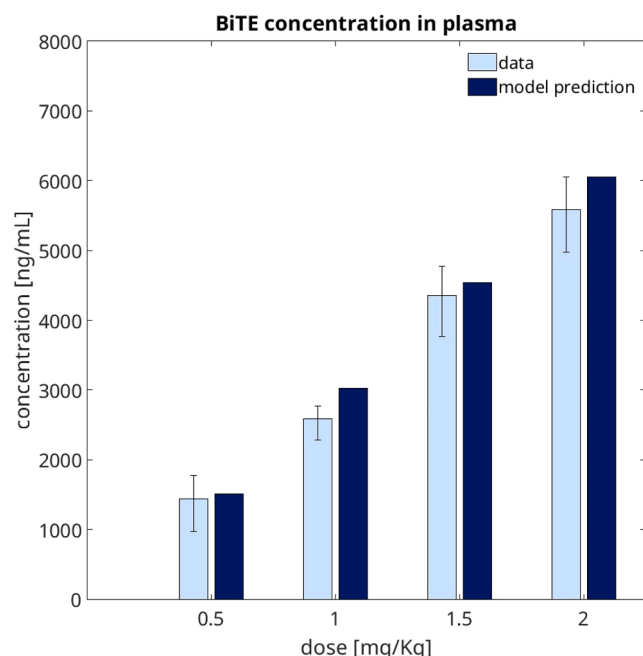


Figure 5. Validation results for the short-chain model

The model was tested to reproduce the plasma concentration of BiTEs 24 h after the administration of four different doses of mRNA, as reported in Huang et al. (2023).²² The mean and uncertainty of the experimental data are represented in light blue, while the *in silico* predicted quantities are shown in dark blue.

nanoparticle, warranting further investigation into a potential change of scenery.

As an example of possible applications of our model, we estimated the optimal mRNA dose needed to achieve a specific antibody concentration within a given time window to ensure therapeutic efficacy (Figure 7A). The *in-silico* simulation aimed to determine the dosage required to observe a minimum concentration of 0.5 ng/mL in the blood 96 h post-administration. Additionally, the model can estimate clearance windows, providing insights into the necessity for subsequent interventions. The model can simulate various therapeutic protocols in the context of the maximum tolerated dose (MTD) (Figure 7B). Assuming an MTD of 1.5 mg/kg, we investigated the optimal therapeutic protocol. The model suggests that if the goal is to maximize antibody concentration in the blood, a single injection might be more effective, whereas dividing the dose into multiple injections might better maximize the coverage window.

Finally, given that our model extends a literature PBPK platform equipped with various calibrations for diverse preclinical settings, future efforts could expand its application to support the translational development of BiTEs. Although the current study does not include data from multiple species, leveraging the work of Shah and Betts (2012)²⁸ would allow for the translation of results across species and toward human applications. It is important to

note, however, that leveraging our model in clinical settings needs careful consideration of several physiological and metabolic differences between species (e.g., glomerular filtration in the kidneys,²⁹ the permeability of the blood-brain barrier,³⁰ etc.), making the translation from mouse to human achievable, but not straightforward. We plan to explore this possibility in future extensions of our model.

While acknowledging the strengths of our model, we must also recognize certain limitations. The prediction accuracy is most reliable within a confined time frame ($t < 24$ h), providing a conservative estimation of the area under the curve in blood. Nonetheless, the projected antibody concentrations remain consistent with the variance of experimental findings. Moreover, the validation process relies predominantly on data from a single experimental study, underscoring the need for comparisons across multiple studies under consistent experimental conditions to better assess the reliability of the model. In this regard, the experiments presented in several recent studies^{31–33} could be leveraged in the future. Notably, the quantitative data reported in all of these studies are still PK profiles of the antibody in plasma, and while this measure provides great insight into the kinetics of the antibody, it does not report much information on the actual concentration of BiTE inside the different organs. Indeed, while one of the potential strengths of this model would be to predict the concentration of BiTE inside remote tissues, additional concentration-time profile data specific for each organ taken into consideration is needed in order to validate this prediction. Moreover, antibody PK profiles do not provide information on the destiny of LNPs and mRNA molecules, making it difficult to predict the fate of these drug components.

One promising point of extension of our work is to support different LNP tropisms in our model. While liver targeting is not inherently limiting—given that LNPs naturally exhibit liver tropism—the potential to deliver LNPs to other organs is crucial for treating a broader range of diseases.³⁴ This consideration aligns with ongoing research efforts focused on developing lipid formulations for non-hepatic delivery, particularly to the lungs, which is gaining significant attention in recent literature.^{35–37}

When additional data—which may give a quantitative dimension to recent *in-vivo* imaging studies^{38,39}—on LNP and mRNA trafficking at the whole-body scale become available to support a less parsimonious model description, future expansions of our model may consider a more detailed representation of the biodistribution of the therapeutic, in agreement with other recent research efforts in the PBPK area^{40–42} that could also contain the description for diverse tropisms and affinities.

Despite this, the reliability of our model predictions is supported by the fact that the novel equations introduced in the PBPK platform are limited to mRNA trafficking and hepatic translation, while the trafficking of the translated therapeutics at the whole-body scale relies

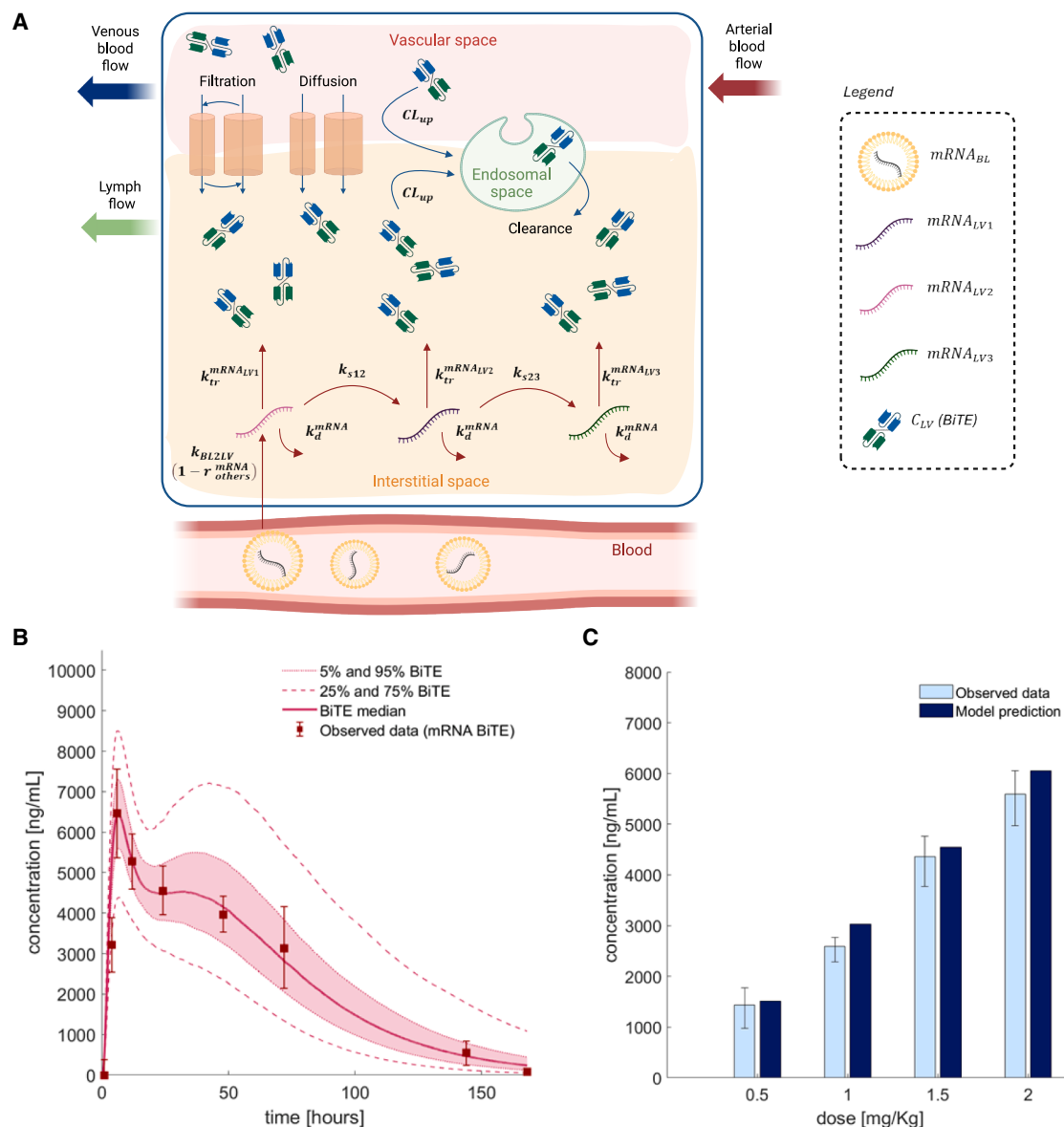


Figure 6. Schematic representation, training and validation results for the long-chain model

(A) Schematic representation of the tissue level dynamics modeled in the liver for the long-chain model (LCM). The mRNA injected into the blood is transferred to the liver and its translation takes place in the interstitial space through three mRNA states that contribute to BiTE production. Similarly to the SCM, when BiTEs are produced, they are transported throughout the body via the PTM. (B) LCM training results on plasma time series of the mRNA-encoded BiTE reported in Huang et al. (2023).²² Uncertainty of the mRNA-based output was estimated by performing a virtual population study. Solid lines represent the median of the concentration in the plasma of BiTEs, dot lines represent the first and the third quartile and dashed lines represent the 5% and 95% of estimated values in the two experimental settings, respectively. (C) LCM validation results: the model was tested to reproduce the plasma concentration of BiTEs 24 h after administration of four different doses of mRNA, as reported in Huang et al. (2023).²² The mean and uncertainty of the experimental data are represented in light blue, while the *in silico* predicted quantities are shown in dark blue. Created in BioRender. Marchetti, L. (2025) <https://BioRender.com/a19j801>.

on the work of Li and Shah (2019),¹⁷ which has been employed in several modeling efforts (such as studies by Jairam et al.,⁴³ Li et al.,⁴⁴ and Hu and D'Argenio⁴⁵), and is based on the seminal work of Shah and Betts (2012),²⁸ underscoring its credibility and utility in this field.

With the availability of more diverse experiments that consider various delivery methods, it could also be feasible to model different routes of administration, harnessing the intrinsic modularity of PBPK models. Developing such a modular model could further accelerate and enhance the precision of LNP-mRNA therapeutic development.

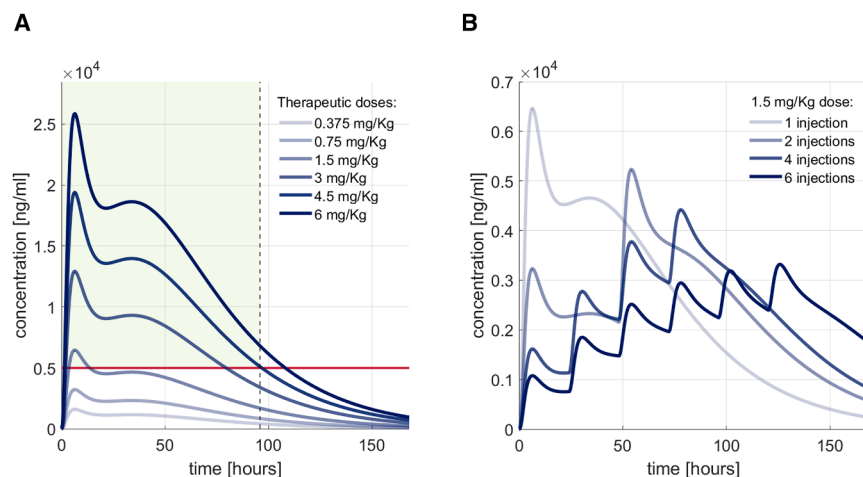


Figure 7. Application examples of our models

(A) An example of the model capability to support the dose-finding process. Model predictions suggest that a minimum dose of 4.5 mg/kg of LNP-mRNA product is required to ensure a minimum concentration of 0.5 ng/mL (solid red line) in the blood 96 h post-administration (green-colored region). (B) An example of the model capability to support the schedule-finding process in the context of a maximum tolerated dose (MTD), assumed here to be 1.5 mg/kg. Results indicate that a single injection might be more effective for maximizing antibody concentration (light-blue line), while dividing the dose into multiple injections might better maximize the coverage window (darker blue lines).

Additionally, as the topic of off-target effects is undoubtedly central to the study of the pharmacokinetics of therapeutics, we have to underline that the model is not yet able to provide insights into this matter. In fact, while the off-target binding is a recurrent issue in the development of antibodies,⁴⁶ it can be assessed only by experiments and modeled afterward.

Moreover, we would like to emphasize that in this initial study, we opted for a simplified, phenomenological description of the biological processes involved, focusing primarily on the early phases of the dynamics, which does not account for the interplay between the administered therapeutics and innate and adaptive immune responses. This decision was driven by the limited experimental data available at the time, which did not allow for the calibration of more complex immune response terms. Additionally, the relatively short temporal range of the considered studies further justifies the use of this phenomenological approach. We acknowledge that the involvement of the immune response plays a significant role, particularly in cases where long-term repeated doses may be required for disease management, as in cancer treatment;^{47,48} however, describing its vast complexity⁴⁹ was out of the scope of this work. Thus, the predicted dose-repeated time series presented in Figure 7B are not subjected to any repeated dosing tolerance. However, new quantitative data on, for example, the known IgM production^{50,51} (and therefore IgG for long-term immunity) or the activation of inflammatory pathways upon pinocytosis⁵² would allow us to make more precise modeling choices to reflect tolerance.

Throughout the development of our platform, significant consideration was given to assessing the identifiability of parameters, aiming to ascertain its efficacy not only in dose and schedule optimization but also in comparative analyses of various products. Our findings reveal that both the SCM and LCM exhibit structural local identifiability. As commonly observed in models of this nature, parameter identifiability tends to be localized, necessitating further investigation to obtain estimates conducive to the fitting process. Therefore, at this modeling stage, we refrain from discussing the estimated

values of rates. However, this constitutes an area of focus for future research endeavors aimed at elucidating the properties of the molecules under study within our platform.

Efforts are underway to enhance the versatility of our platform to accommodate a broader range of proteins for trafficking analysis, including the incorporation of the neonatal Fc receptor (FcRn), acknowledging its substantial impact on the longevity of IgG-like antibodies and their therapeutic effectiveness through extended half-life and continual functionality against pathogens.^{18,19} Validation of this expanded platform should be conducted using additional data, with a concerted effort to determine the minimal conditions necessary for achieving global identifiability.

In conclusion, this study underscores the potential of combining mRNA-based technology with BiTEs as a promising strategy in cancer therapy. Utilizing a comprehensive multiscale PBPK model, we accurately predicted the distribution and pharmacokinetics of mRNA-encoded BiTEs across various organs and tissues. This model, validated with preclinical data, captures the intricate relationship between mRNA dosage and protein production, enabling the exploration of drug distribution at both molecular and whole-body levels. By supporting informed decision-making in therapeutic interventions, particularly in optimizing dosing strategies and predicting responses across diverse physiological contexts, this framework lays the groundwork for advancing mRNA-based therapies toward clinical applications.

MATERIALS AND METHODS

Data

Most of the physiological parameters for the two-pore PBPK model are taken from Li and Shah (2019),¹⁷ whose values are based on Shah and Betts (2012).²⁸ These studies were selected due to their versatility in handling proteins of different sizes and their applicability across species, enabling the characterization of monoclonal antibodies across various organisms. Notably, the dataset used in Shah and Betts (2012)²⁸ was compared to another dataset in Sepp et al. (2019),⁵³

enhancing the validation of the data. Our model is designed to predict the trafficking of proteins of different sizes, with a specific focus on the BiTE protein, which has an approximate size of 55 kDa. Any parameter dependent on the size of the therapeutic, such as those regarding the two-pore dynamics and the glomerular filtration rate, has been calculated according to this size value.

The training and validation datasets employed to develop our model are described in Huang et al. (2023),²² who tested a BiTE-based mRNA treatment in mice. After comprehensive *in vitro* testing of the B7H3×CD3 BiTE protein, the authors developed ionizable LNPs encapsulating mRNA encoding B7H3×CD3 BiTE. *In vivo* experiments have been conducted in 8–10 weeks old immunodeficient tumor-bearing mice. The mice were injected with single i.v. doses of 120 µg (6 mg/kg) for BiTE and 30 µg (1.5 mg/kg) for mRNA-BiTE. Both BiTE and mRNA-BiTE were administered intravenously, and the concentration of BiTE in serum was measured over 7 days after i.v. administration (1, 4, 6, 12, 24, 48, 72, 144, and 168 h). In addition, increasing doses of mRNA-BiTE (0, 0.5, 1, 1.5, and 2 mg/kg) were administered to different mice, and the corresponding concentration of BiTE in serum was assessed at a single time point (24 h) for each mouse. The data of such an escalation dose experiment have been used for validating our model. We specify that the PBPK model by Li and Shah (2019),¹⁷ which forms the foundation of our work, utilizes plasma concentrations. Although the experimental data from Huang et al.²² are reported in serum, we can still calibrate the model using these data. This approach is supported by the literature, which indicates that serum and plasma concentrations can be considered equivalent for PK modeling when the drug does not bind extensively to clotting factors or proteins since BiTEs (and antibodies in general) have extremely precise binding affinities.⁵⁴

Model implementation

The model was implemented using MATLAB (R2022b) and the simulations were executed on a system equipped with an Intel Core i7-8700T processor, CPU 2.40 GHz, with 16.00 GB installed RAM. Model simulations were computed by relying on the ODE solver *ode15s*, configured with an error tolerance of 10^{-12} .

For model training, we used the MATLAB function *createOptimProblem* to define an optimization problem structure based on a cost function that calculates the relative deviations at specific time points (1, 4, 6, 12, 24, 48, 72, 144, and 168 h) using the L2 norm and we employed the *Multistart* object to facilitate a global search during parameter estimation. The algorithm performed 50 runs of the *lsqnonlin* algorithm, with an error tolerance of 10^{-12} .

For the fitting results shown in Figures 4 and 6, we computed a virtual population of 1,000 individuals by varying the estimates of the parameters introduced for the mRNA layer and the endothelial pinocytosis rate. These parameters were varied according to a state-of-the-art approach using a normal distribution, with the mean set to the fitted value and a relative standard deviation of 0.2.

The identifiability analysis of the model was performed using GenSSI,⁵⁵ a toolbox for MATLAB computing language. The analysis confirmed the local identifiability of all model parameters reported in Table 1.

DATA AND CODE AVAILABILITY

All data supporting the findings described in this manuscript are available in the paper and in the list of references. The MATLAB source code of the models is available at <https://github.com/cosbi-research/PBPKmRNABiTE>.

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AUTHOR CONTRIBUTIONS

Conceptualization, G.F., L.L., and L.M. Data curation, G.F., E.C., L.L., E.P., S.G., and L.M. Formal analysis, G.F., E.C., L.L., E.P., S.G., and L.M. Funding acquisition, L.M. Investigation, G.F., E.C., L.L., E.P., S.G., and L.M. Methodology, G.F., E.C., E.P., S.G., and L.M. Project administration, L.M. Resources, L.M. Software, G.F., E.C., L.L., E.P., S.G., and L.M. Supervision, L.M. Validation, G.F., E.C., L.L., E.P., S.G., and L.M. Visualization, G.F., E.C., L.L., E.P., S.G., and L.M. Writing – original draft, G.F., E.C., L.L., E.P., S.G., E.J.C., L.D., N.Z., and L.M. Writing – review & editing, G.F., E.C., L.L., E.P., S.G., E.J.C., L.D., N.Z., and L.M.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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